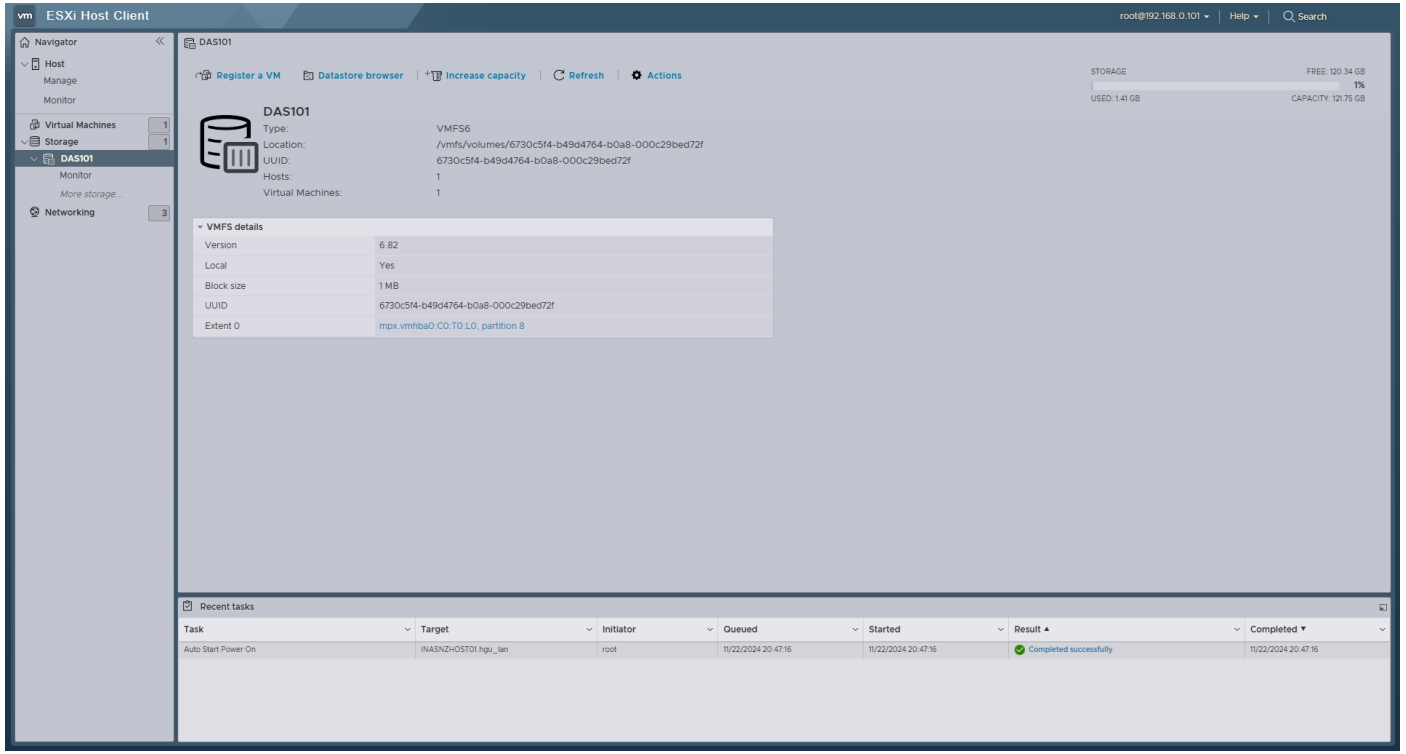
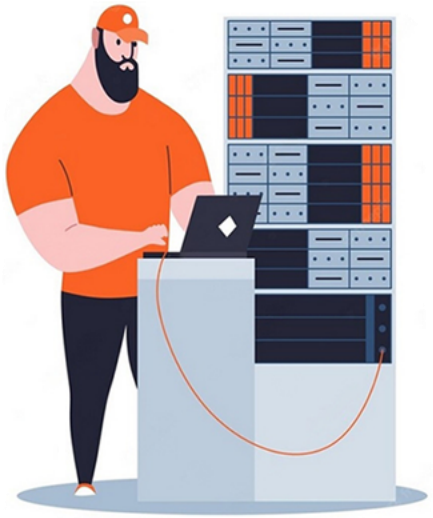
AGENDA

About File System (VMFS) in VMware ESXi 8.0

Version of VMFS in VMware ESXi 8.0

Features of VMFS in VMware ESXi 8.0





VMware ESXi
HYPERVISOR

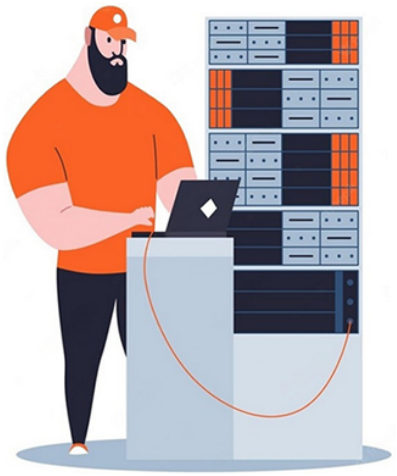
ABOUT FILE SYSTEM IN VMware ESXi 8.0

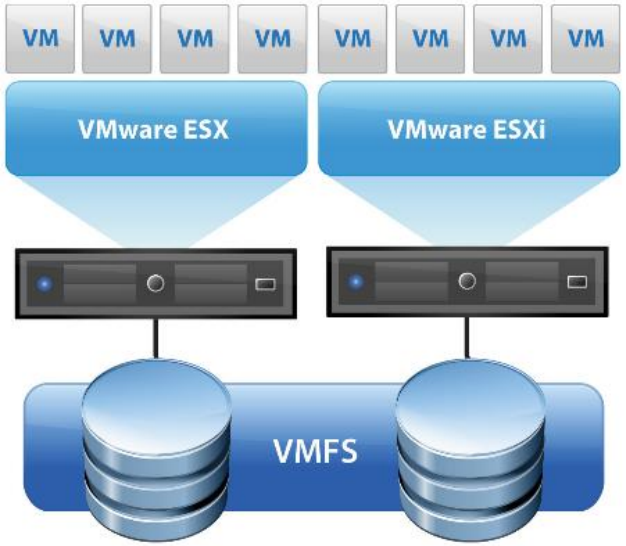


FILE SYSTEM IN VMware ESXi 8.0

To store virtual disks, ESXi uses datastores. The datastores are logical containers that hide specifics of physical storage from virtual machines and provide a uniform model for storing the virtual machine files. The datastores that you deploy on block storage devices use the native vSphere Virtual Machine File System (VMFS) format. It is a special high-performance file system format that is optimized for storing virtual machines.

You can increase the capacity of the datastore while the virtual machines are running on the datastore. This ability lets you add new space to your VMFS datastores as your virtual machine requires it. VMFS is designed for concurrent access from multiple physical machines and enforces the appropriate access controls on the virtual machine files.





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VERSION'S OF VMFS IN VMware ESXi 8.0



VERSION'S of VMFS IN VMware ESXi 8.0

There are four versions of VMFS, corresponding with ESX Server product releases.

VMFS1 was used by ESX Server v1.x. It did not feature the cluster file system properties and was used only by a single server at a time. VMFS1 is a flat file system with no directory structure.

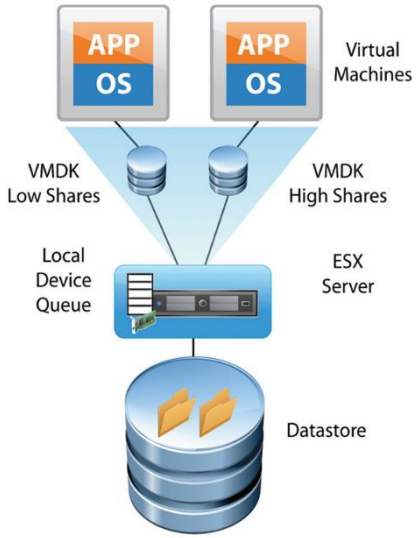
VMFS2 is used by ESX Server v2.x and (in a limited capacity) v3.x. VMFS2 is a flat file system with no directory structure.

VMFS3 is used by ESX Server v3.x and vSphere 4.x. It introduces directory structure in the file system.

VMFS5 is used by vSphere 5.x. It raises the extent limit to 64 TB and the file size limit to 62 TB, though vSphere versions earlier than 5.5 are limited to VMDKs smaller than 2 TB.

VMFS-L (VMFS6) is the underlying file system for VSAN Leaf level VSAN objects reside directly on VMFS-L volumes that are composed from server side direct attached storage (DAS). File system format is optimized for DAS. Optimization include aggressive caching with for the DAS use case, a stripped lock down lock manger and faster formats.





VMware ESXi
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VERSION'S of VMFS IN VMware ESXi 8.0



Features and Functionalities	VMFS5	VMFS6
Access for ESXi hosts version 6.5 and later	Yes	Yes
Access for ESXi hosts version 6.0 and earlier	Yes	No
Datastores per host	512	512
512n storage devices	Yes	Yes (default)
512e storage devices	Yes. Not supported on local 512e devices.	Yes (default)
4Kn storage devices	No	Yes
Automatic space reclamation	No	Yes
Manual space reclamation through the esxcli command. See Manually Reclaim Accumulated Storage Space.	Yes	Yes
Space reclamation from guest OS	Limited	Yes
GPT storage device partitioning	Yes	Yes
MBR storage device partitioning	YesFor a VMFS5 datastore that has been previously upgraded from VMFS3.	No
Storage devices greater than 2 TB for each VMFS extent	Yes	Yes
Support for virtual machines with large capacity virtual disks, or disks greater than 2 TB	Yes	Yes
Support of small files of 1 KB	Yes	Yes
Default use of ATS-only locking mechanisms on storage devices that support ATS. See VMFS Locking Mechanisms.	Yes	Yes
Block size	Standard 1 MB	Standard 1 MB
Default snapshots	VMFSsparse for virtual disks smaller than 2 TB.SEsparse for virtual disks larger than 2 TB.	SEsparse
Virtual disk emulation type	512n	512n
vMotion	Yes	Yes
Storage vMotion across different datastore types	Yes	Yes
High Availability and Fault Tolerance	Yes	Yes
DRS and Storage DRS	Yes	Yes
RDM	Yes	Yes

VMware ESXi
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FEATURE OF VMFS IN VMware ESXi 8.0



FEATURE OF VMFS IN VMware ESXi 8.0

- Allows access by multiple ESX/ESXi servers at the same time by implementing per-file locking. SCSI reservations are only implemented when logical unit number (LUN) metadata is updated (Example:- file name change, file size change, etc.)
- Add or delete an ESX server from a VMware VMFS volume without disrupting other ESX servers.
- With ESX/ESXi4, VMFS volumes also can be expanded using LUN expansion
- Optimize virtual machine I/O with adjustable volume, disk, and file and block sizes.
- Recover virtual machines faster and more reliably in the event of server failure with Distributed Journaling.

